

LAG-1 Session 1: Bergman Dirac Skeleton + Hua-coord Candidate

Lyra

2026-05-17

LAG-1 Session 1: Skeleton + Hua-coord Candidate

Goal of this session

Set up the Bergman Dirac operator on D_{IV}^5 in concrete terms: - Choose representation (real-10 manifold $\mathbb{H}^5$ $Cl(10)$ Clifford algebra $\mathbb{H}^5$ 32×32 Dirac spinors, OR complex-5 $\mathbb{H}^5$ $Cl_C(5)$ with 8-dim complex spinors) - Write the candidate operator $\gamma_B^\mu \partial_\mu$ on a generic $z \in D_{IV}^5$ in Hua coordinates - Verify the Lichnerowicz-formula dimensional consistency (D^2 has the right number of independent components vs Laplacian + scalar curvature)

Session 1 is FRAMEWORK + DIMENSIONAL CHECK. Sessions 2-3 do the explicit matrix construction.

Choice of representation

D_{IV}^5 has: - Complex dim = $n_C = 5$ - Real dim = rank $\cdot n_C = 10$

The Dirac operator can be set up two ways:

Real-10 approach: D_{IV}^5 as a real 10-manifold; Clifford algebra $Cl(10, 0)$ (or $Cl(8, 2)$ for the signature 5+5 if we treat the Hermitian metric as split signature). Spinors are $2^5 = 32$ -dimensional. Dirac matrices are 32×32 .

Complex-5 approach: D_{IV}^5 as a complex 5-manifold; use the holomorphic-anti-holomorphic decomposition. Dirac becomes $\partial + \bar{\partial}^*$ (the Dolbeault operator) acting on $(0, *)$ -forms tensored with the spinor bundle. Spinors are $2^{n_C} = 32$ -dim too, but split as $16 + 16$ (chiral pair).

We pick complex-5 / Dolbeault approach because: 1. D_{IV}^5 is naturally a Hermitian symmetric space; complex structure is preferred 2. Chirality decomposition matches T1947's chirality + CP structure 3. Holomorphic discrete series (Wallach K-types) directly give Dirac spectrum

Setup in Hua coordinates

Coordinates: $z = (z_1, \dots, z_5) \in \mathbb{C}^5$ with D_{IV}^5 defining inequalities

$$1 - 2|z|^2 + |z \cdot z|^2 > 0, \quad |z|^2 < 1$$

(Recall from T2328: real-form locus $M(D_{IV}^5) = \{z = \bar{z}\} = \text{open 5-ball in } \mathbb{R}^5$.)

Bergman metric: $g_{i\bar{j}} = \partial_i \partial_{\bar{j}} \log K_B(z, \bar{z})$ where $K_B = c \cdot D(z, \bar{z})^{-(n_C+2)/2}$ (T2334).

For the Bergman Dirac on D_{IV}^5 :

$$D_B = \gamma^\mu (\partial_\mu + (1/4) \omega_\mu^{\{ab\}} \gamma_a \gamma_b)$$

where ω is the spin connection of the Bergman metric, γ^μ are Dirac matrices for the Hermitian inner product.

Hua-coordinate candidate (skeleton)

In Hua coordinates with the Bergman metric, the leading-order Dirac action is:

$$D_B = \gamma^z_i \partial_{\{z_i\}} + \gamma^{\bar{z}}_i \partial_{\{\bar{z}_i\}} + 0(\bar{z} \cdot z)$$

where the $\gamma^z_i / \gamma^{\bar{z}}_i$ pair satisfies the Clifford anti-commutation relations:

$$\{\gamma^z_i, \gamma^{\bar{z}}_j\} = 2 g^{\{ij\}} = 2 \text{ (Bergman metric inverse)}$$

For D_{IV}^5 : $g^{\{ij\}}(0) = \delta^{\{ij\}}$ (at origin); higher-order corrections involve the Hermitian + bilinear terms in $D(z, \bar{z})$.

Lichnerowicz formula dimensional check

The Lichnerowicz identity says:

$$D_B^2 = \nabla * \nabla + R/4$$

where R is the scalar curvature.

For D_{IV}^5 , the scalar curvature in Bergman metric is:

$$\begin{aligned} R &= -n_C \cdot (n_C + 2) \text{ (negative Einstein, sign for Hermitian symmetric domain)} \\ &= -5 \cdot 7 = -35 = -n_C \cdot g \end{aligned}$$

So $R/4 = -35/4$ (in units where Bergman metric scale is normalized).

Dimensional consistency target: D_B^2 acting on a 32-dim spinor on a 10-dim manifold has 32... independent operator components. The Laplacian $\nabla * \nabla$ has the same count. The $R/4$ term is a scalar multiple of identity. The Lichnerowicz formula has the right number of degrees of freedom on both sides.

BST integer in the scalar curvature: $R = -n_C \cdot g = -35$. Both n_C and g are BST primaries. The “35” itself is $N_c \cdot n_C \cdot g / N_c = n_C \cdot g$ (matches Stark-Heegner $d=35$ isn’t on the list; but $35 = n_C \cdot g$ is structural).

What this session produces

1. This document — framework + representation choice + Hua-coord skeleton
2. Toy 3002 — dimensional consistency check + scalar curvature $R = -n_C \cdot g$ verification + BST integer structure
3. T2339 registry entry — partial first-session result

Open after Session 1

- Session 2: explicit $32 \times 32 \gamma^\mu$ matrices for the complex-5 Dolbeault approach
- Session 3: spin connection ω from Bergman metric (this is where the geometry enters)
- Sessions 4-8 as scoped

Why this matters

This is the first concrete piece of LAG-1. From Session 2 onward we have actual matrix computations to do; Session 1 just sets up the framework so we know what we’re computing.

Net Session 1 effort: ~45 min (this doc + toy).

— Lyra, 2026-05-17 ~15:50 EDT